



Evaluation of Probiotic Properties of Novel Brazilian *Lactiplantibacillus plantarum* Strains

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Abstract

Beneficial effects of *Lactiplantibacillus plantarum* strains have been widely reported. Knowing that the effects of probiotic bacteria are strain-dependent, this study aimed to characterize the probiotic properties and investigate the gastrointestinal protective effects of nine novel *L. plantarum* strains isolated from Bahia, Brazil. The probiotic functionality was first evaluated in vitro by characterizing bile salt and acidic tolerance, antibacterial activity, and adhesion to Caco-2 cells. Antibiotic resistance profile, mucin degradation, and hemolytic activity assays were also performed to evaluate safety features. In vivo analyses were conducted to investigate the anti-inflammatory effects of the strains on a mouse model of 5-Fluorouracil-induced mucositis. Our results suggest that the used *L. plantarum* strains have good tolerance to bile salts and low pH and can inhibit commonly gastrointestinal pathogens. Lp2 and Lp11 strains also exhibited high adhesion rates to Caco-2 cells (13.64 and 9.05%, respectively). Phenotypical resistance to aminoglycosides, vancomycin, and tetracycline was observed for most strains. No strain showed hemolytic or mucolytic activity. Seven strains had a protective effect against histopathological and inflammatory damage induced by 5-FU. Gene expression analysis of inflammatory markers showed that five strains upregulated interleukin 10 (*IL10*), while four downregulated both interleukin 6 (*IL6*) and interleukin 1b (*IL1b*). Additionally, all strains reduced eosinophilic and neutrophilic infiltration; however, they could not prevent weight loss or reduced liquid/food intake. Altogether, our study suggests these Brazilian *L. plantarum* strains present good probiotic characteristics and safety levels for future applications and can be therapeutically adjuvant alternatives to prevent/treat intestinal mucositis.

Keywords Lactic acid bacteria · Probiotic potential · Stress tolerance · Antimicrobial activity · Anti-inflammatory effect · Intestinal mucositis

Introduction

Probiotics are “live microorganisms which, when administered in adequate amounts, confer a health benefit on the host” [1]. Currently, the use of these microorganisms

for human consumption has received increasing attention due to the discovery of new scientific evidence regarding their functional properties and beneficial effects on the host [2]. Most probiotic strains belong to the Lactic Acid Bacteria (LAB) group [3]. Among them, *Lactiplantibacillus*

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plantarum (previously classified as *Lactobacillus plantarum*) is one of the most common microorganisms found in the gastrointestinal tract (GIT) of humans and other animals. This species is involved in crucial biological processes, such as intestinal food fermentation, metabolites production, epithelial barrier function enhancement, and pathogen colonization inhibition [4–7].

Some *L. plantarum* strains are known to have beneficial effects, such as the production of antimicrobial compounds [8, 9], cholesterol-lowering levels, anti-obesity effect [10–12], improvements in psychological disorders such as depression and anxiety [7], and prevention of colitis and other intestinal diseases [13–15]. Nevertheless, the effects of these bacteria are strain-dependent, and the benefits for a specific strain cannot be extrapolated to others within the same species [16]. Thus, evaluating the potential probiotic effects of individual strains and their beneficial effects on different pathological conditions is essential.

Several studies have shown the beneficial effect of LAB strains on mitigating chemotherapy-induced intestinal mucositis [17–20], which is an ulcerative inflammation of the small intestine mucosa of patients undergoing cancer treatment, mainly chemotherapy (e.g., 5-Fluorouracil) [21, 22]. The clinical manifestations of 5-FU-induced mucositis include nausea and vomiting, abdominal pain, weight loss, and diarrhoea. These manifestations directly affect patients' quality of life and often lead to discontinuation and, consequently, reduction in the efficacy of the chemotherapy treatment [23].

Knowing that intestinal microbiota is significantly altered by chemotherapeutic treatments and contributes to the progression and severity of mucositis [24], the use of probiotics could be a promising therapeutic option to ameliorate this oncological problem, due to their ability to regulate the intestinal microbiota and reduce intestinal inflammation [17, 25, 26]. Currently, palliative therapy is the only treatment proposed for chemotherapy-induced mucositis. Besides being a good candidate, studies involving *L. plantarum* strains to prevent or improve mucositis symptoms are still lacking [27, 28]. Thus, through this study, we characterized the probiotic property and safety aspects related to nine novel *L. plantarum* strains isolated from different sources in Brazil and their protective and anti-inflammatory activity in a mouse model of 5-FU-induced intestinal mucositis.

Methods

Bacterial Strains, Growth Conditions, and Identification by MALDI-TOF Biotyper®

The bacterial strains analyzed in this work were isolated from different sources in Bahia, Brazil. They were catalogued

in the collection of the Laboratory of Applied Microbiology and Public Health (LAMASP—State University of Feira de Santana—UEFS, Bahia, Brazil) (Table 1). All strains were cultivated in MRS broth or agar (de Man, Rogosa, and Sharpe) (Kasvi, São José dos Pinhais, Brazil) at 37 °C for 12 h and 24 h, respectively.

For microbiological identification, the overgrown strains were plated using a sterile plastic loop on MRS agar plates (Kasvi, São José dos Pinhais, Brazil) and incubated at 37 °C for 24 h. The colony identification was performed by MALDI-TOF Biotyper® Mass Spectrometry (Bruker Daltonics, Billerica, MA, USA) according to the manufacturer's instructions.

In Vitro Assays

Gastric Juice Tolerance

The tolerance to acidic conditions of different isolates was evaluated according to Tokatli et al. [29]. The strains were grown overnight in MRS broth (Kasvi, São José dos Pinhais, Brazil), washed twice with PBS 0.1 M (37 mM NaCl, 2.7 mM KCl, 8 mM Na₂HPO₄, and 2 mM KH₂PO₄, pH 7.0), and suspended in 1 mL of artificial gastric juice (NaCl 2 g/L, pepsin from porcine gastric mucosa 3.2 g/L) (Sigma-Aldrich, St. Louis, MO, USA), with pH adjusted to 2.5. Afterwards, the solution was incubated at 37 °C for 3 h under 150 rpm agitation, washed with PBS 0.1 M, and suspended in 2 mL of MRS broth (Kasvi, São José dos Pinhais, Brazil). A total of 200 µL of each strain culture was inoculated in a 96-well flat-bottom plate and incubated in a Microplate Spectrophotometer (Multiskan Spectrum—Thermo Scientific) at 37°C for 12 h without agitation. The absorbance was measured at 600 nm every hour for 12 h to obtain the growth kinetics curves. The percentage of inhibition at low pH was calculated by the difference between the area under the control curve and the test's growth curve. This experiment was performed in triplicate.

Table 1 *L. plantarum* strains identification used in this study and their respective source

Source	Strain ID
Artisanal butter- Feira de Santana—Bahia, Brazil	Lpl2
Manioc—Feira de Santana—Bahia, Brazil	Lpl3
Sugarcane juice – Feira de Santana—Bahia, Brazil	Lpl4
Cow's milk – Farm in Ruy Barbosa—Bahia, Brazil	Lpl9
Beetroot—Feira de Santana—Bahia, Brazil	Lpl11
Pineapple – Feira de Santana—Bahia, Brazil	Lpl14
Cow's milk – Feira de Santana—Bahia, Brazil	Lpl17
3 days newborn faeces– Feira de Santana—Bahia, Brazil	Lpl18
Foal faeces- Farm in Ruy Barbosa—Bahia, Brazil	Lpl19

Bile Salt Tolerance

Bile salt tolerance assay was carried out according to Walker and Gilliland [30] with some modifications. Briefly, each overnight-grown culture was diluted in MRS broth (control) or MRS broth containing 0.3% (m/v) of bile salt Oxgall (Sigma-Aldrich, St. Louis, MO, USA) until it reached a 0.4 optic density at 600 nm. The cultures were distributed in a 96-well-plate and were incubated at 37 °C in a spectrophotometer (Multiskan Spectrum—Thermo Scientific). The absorbance was measured at 600 nm every hour for 12 h to obtain growth kinetics curves. The difference between the area under the control curve and the test's growth curve was calculated to assess the bile salts inhibition percentage. This experiment was performed in triplicate.

Antibacterial Activity

The spot-on-lawn antimicrobial assay was performed to evaluate the antibacterial activity of *L. plantarum* strains, as previously described by Touré et al. [31], with some modifications. For this assay, six bacteria related to human gastrointestinal diseases were used as indicator strains: *Escherichia coli* (ATCC® 11,775), *Shigella sonnei* (ATCC® 9290), *Salmonella enterica* serovar Typhimurium (ATCC® 14,028), *Salmonella enterica* serovar Typhi (ATCC® 70,093), *Enterococcus faecalis* (ATCC® 19,433), and *Listeria monocytogenes* (ATCC®15,313). Briefly, 5 µL of overnight-grown *L. plantarum* cultures were spotted on MRS agar (Kasvi) and incubated for 24 h at 37 °C. Pathogenic strains were grown overnight in Brain–Heart Infusion (BHI) agar (Sigma-Aldrich, St. Louis, MO, USA) at 37 °C. Afterwards, the grown colonies were inactivated with 1 mL chloroform (Labsynth, São Paulo, Brazil) for 30 min. Plates were left open for another 30 min to remove any residual chloroform. Subsequently, 10 µL of each indicator strain (final concentration of 9 log CFU/mL) was inoculated into semi-solid 1.0% BHI agar and then placed on the top of the MRS agar layer. Then, the plates were incubated for 24 h at 37 °C, and the inhibition zone around the colonies was measured with a millimeter ruler. Antibacterial activities were classified as strong (diameter ≥ 20 mm), moderate (20 mm < diameter > 10 mm) and weak (diameter ≤ 10 mm) [31]. This test was performed in duplicate.

Adhesion to Caco-2 Cells

The adhesion ability of the isolates was carried out using the human intestinal epithelial Caco-2 cell line (ATCC 2102-CRL), according to Nami et al. [32], with some modifications. The cells were cultured in complete Dulbecco's Modified Eagle Medium (DMEM)

(Sigma-Aldrich), supplemented with 20% (v/v) fetal bovine serum (FBS) (Sigma-Aldrich) lacking antibiotics and were incubated at 37 °C, 5% CO₂ until reaching 80–85% confluence. The media was changed every 2–3 days. Then, cells were detached with 3 mL of 0.25% trypsin–EDTA solution (Sigma-Aldrich) for 3–4 min at 37 °C and counted in suspension using a Neubauer Chamber.

For adhesion assay, Caco-2 cells were seeded in each well of a 24-well tissue culture plate (Corning Inc., Corning) at a concentration of 1 × 10⁵ cells/mL and incubated at 37 °C in 5% CO₂ for 21 days until differentiation. Medium change was performed every 48 h and 24 h before the assay. *L. plantarum* strains were grown overnight in MRS broth (Kasvi) and plated into MRS agar plates (Kasvi) after serial dilutions to obtain the exact number of initial bacteria used in the assay. 1 mL of the same overnight culture was harvested by centrifugation for 2 min at 12,000 rpm, suspended in DMEM without FBS (Sigma-Aldrich), added to each well separately, and incubated at 37 °C for 2 h in 5% CO₂. Non-adhered bacteria were withdrawn from the wells by washing 5 times with 1 mL of PBS 1 M each time. The Caco-2 cells were detached by treatment with 0.5 mL of Trypsin–EDTA 0.25% solution (ThermoFisher, Waltham, MA, USA) for 10 min at 37 °C. After that, the detached cells, including bound bacterial strains, were plated after serial dilutions on MRS agar plates (Kasvi). The colonies were counted after 48 h of incubation at 37 °C. The adhesion was calculated as the percentage of adhered bacteria (B1) to total bacteria (B0), as follows: % adhesion = (B1/B0) × 100. The adhesion test was performed in triplicate.

Antibiotic Susceptibility Testing

Antibiotic susceptibility was determined by the disc diffusion method, according to the Clinical and Laboratory Standards Institute (CLSI) guidelines [33], with some modifications. 50 µL of overnight-grown cultures were plated in MRS-agar plates. Afterwards, the antibiotic disks (Cecon®, São Paulo, Brazil) were spotted on the agar surface. The following antibiotics were evaluated with the respective standard concentration: clindamycin (2 µg), erythromycin (15 µg), gentamicin (10 µg), streptomycin (10 µg), tetracycline (30 µg), vancomycin (30 µg), ampicillin (10 µg), and chloramphenicol (30 µg). After plate incubation at 37 °C for 24 h, inhibition zone diameters were measured using a millimeter ruler, and results were expressed as resistant or susceptible, according to the interpretative cut-off described by CLSI [33]. The antibiotic susceptibility test was performed in duplicate.

Mucin Degradation Activity

Salmonella enterica serovar Typhi (ATCC 700,931™) and *Salmonella enterica* serovar Typhimurium (ATCC

14,028TM) were used as a positive control (ATCC®, Manassas, Virginia, USA). The assay was performed according to Zhou et al. [34], with some modifications. 10 µL of overnight-grown cultures were spotted in Luria–Bertani (LB) agar plates (1.5% Peptone, 1% yeast extract, 1% NaCl, and 1.5% granulated agar, pH 7) containing 0.3% of hog gastric mucin-type III (Sigma-Aldrich). Afterwards, the plates were incubated at 37 °C for 48 h, stained with 0.1% amido black (Sigma-Aldrich) in 3.5 M acetic acid for 15 min, and subsequently discoloured with 1.2 M acetic acid (Synth). The mucin degradation activity was defined as the presence of a clear lysis zone towards the *L. plantarum* colonies. This test was performed in duplicate.

Hemolytic Activity

The hemolytic activity assay was carried out according to Xu et al. [35], with some modifications. Briefly, 10 µL of overnight-grown cultures were spotted on tryptic soy agar plates (Kasvi) with 5% defibrinated sheep blood and incubated at 37 °C for 48 h. The hemolysis process was analyzed according to the presence of a clear halo (β-hemolysis), a greenish halo (α-hemolysis), or no alterations (γ-hemolysis) around the colonies. This test was performed in duplicate.

In Vivo Evaluation of Anti-Inflammatory Effects of *L. plantarum* Strains

Mouse Handling and Experimental Design

The anti-inflammatory effects of *L. plantarum* strains were evaluated using a mouse model of 5-FU-induced intestinal mucositis. For this, male BALB/c mice (4–6 weeks old, weight 21–24 g) obtained from the Bioterism Center (CEBIO) of the Federal University of Minas Gerais (Belo Horizonte, Minas Gerais, Brazil) were kept in ventilated polycarbonate boxes under regulated temperature (22 °C ± 2) in a 12-h light/dark cycle-controlled room, with ad libitum water and standard chow diet 24 h before experiments. All following procedures were approved by the Institutional Animal Experimental Ethics Committee (CEUA-UFMG, Protocol number 112/2020) and comply with the guidelines for the care and use of laboratory animals as described by the U.S. National Institutes of Health.

Mice were randomly split into eleven groups (n = 5 animals per group): control (CTL), mucositis (MUC), and nine Lpl groups, consisting of mice with 5-FU-induced mucositis which were treated with one of the nine *L. plantarum* (Lpl) strains each. Animals were orally fed daily with MRS broth (CTL and MUC) or MRS with specific *L. plantarum* strains (10⁹ CFU/mL) for 13 days. Bacterial cultures

were renewed daily to avoid decantation, clogging, or contamination. To induce gastrointestinal mucositis, mice (MUC and Lpl groups) received a single intraperitoneal injection of 5-FU (300 mg/kg) (Fauldfluor®, Libbs, São Paulo, Brazil) on the 10th experimental day, as previously reported by Carvalho et al. [36]. Control group (CTL) received intraperitoneal injections of NaCl 0.9% (w/v). 72 h after administering either 5-FU or saline solution, the animals were euthanized by anaesthetic overdose (30 mg/kg of xylazine and 300 mg/kg of ketamine) (Agener União, Embu-Guaçu, Brazil). The small intestine was collected for analysis. The body weight, hydric, and food intake were assessed daily throughout the experimental period.

Histopathological and morphometric analysis

After euthanasia, the small intestine was removed, and its length was measured using a millimetre ruler. Ileum sections (approximately 20% of the distal small intestine) were longitudinally opened, washed with PBS 0.1 M to remove the stools, rolled up, and immersed in 10% buffered formaldehyde solution (Labsynth, São Paulo, Brazil) for tissue fixation. This material was embedded in paraffin, and 4-µm slides sections were mounted on glass slides and stained with hematoxylin and eosin (HE). For morphological examination, ten images of each animal's ileum slides were captured with an optical microscope (Olympus BX41, Tokyo, Japan) with a 20× magnification objective. Images were analyzed using ImageJ 1.51j.8 software (NIH, Bethesda, MD, USA) to measure villus height (VH) and crypt depth (CD). The histopathological analysis was performed blindly by a pathologist according to the scoring system proposed by Soares et al. [37].

Inflammatory Cells Infiltration

The polymorphonuclear cell infiltration (neutrophils and eosinophils) in the intestinal mucosa was determined by myeloperoxidase (MPO) and eosinophil peroxidase (EPO) enzyme activities, respectively [38]. Briefly, 100 mg of ileum tissue was homogenized in 1.9 mL phosphate-buffered saline (PBS 0.1 M, pH 7.4), centrifuged at 12,000 rpm for 10 min at 4 °C, and the pellets were subjected to hypotonic lysis using 1.5 mL of 0.2% NaCl. The osmolality was restored with 1.5 mL of 1.6% NaCl solution with 5% glucose. Subsequently, samples were centrifuged again at 12,000 rpm for 10 min at 4 °C, and the pellet was suspended in 0.5% hexadecyltrimethylammonium bromide (HTAB-Sigma-Aldrich) in PBS 0.1 M. The suspension was homogenized, snap-frozen in liquid nitrogen three times, and centrifuged for 15 min at 12,000 rpm at 4 °C. The resulting supernatant was used in the colourimetric assay to measure enzymatic activities. Absorbance was measured at 492 nm (EPO), and

450 nm (MPO) on a microplate spectrophotometer (Bio-Rad 450 model, Bio-Rad Laboratories, Hercules, CA, USA), and the results were expressed as MPO or EPO arbitrary units (based on absorbance)/100 mg of tissue.

Ileal Cytokines Gene Expression

Ileum sections (1 cm) were stored in RNA later (Sigma-Aldrich) at -20°C until use. Total tissue RNA was extracted using PureLink RNA Isolation Kit (ThermoFisher), and the complementary DNA (cDNA) was synthesized using the High-Capacity cDNA Reverse Transcription (ThermoFisher) kit, according to the manufacturer's instructions. Quantitative PCR (qPCR) was performed using the PowerUp™ SYBR® Green Master Mix (ThermoFisher) to measure mRNA expression levels of specific genes *Il10*, *Il6*, and *Il1b* using the primers described in Table 2. Amplification reactions were performed with 7900HT Fast Real-Time PCR System (Applied Biosystems) under the following conditions: initial denaturation at 95°C for 10 min, 40 cycles of 95°C for 15 seg, annealing/extension at 60°C for 1 min, followed by a dissociation stage for recording the melting curve. The expression of target genes was analyzed by the $2^{-\Delta\Delta\text{Ct}}$ method using a housekeeping gene encoding GAPDH as a normalizer.

Statistical Analysis

The Shapiro Wilk test assessed the data normality. The data were evaluated by one-way analysis of variance (ANOVA) test followed by Tukey's post-test (parametric distribution). Non-parametric data were evaluated by the Kruskal–Wallis test followed by Dunn's post-test. The Student t-test was conducted to analyze the mean differences between the two groups. All data were analyzed using the GraphPad Prism 8.0 software, and the p -value was <0.05 .

Results

Species Identification

The identification by MALDI-TOF Biotyper® classified all strains as belonging to the *L. plantarum* species (score >2.1). The secure genus identification was obtained for all strains,

while four of them (Lpl4, Lpl9, Lpl11, and Lpl14) presented accurate species-level identification.

In Vitro Assays Findings

L. plantarum Strains Tolerate Acidic and Bile Salt Stress

All strains showed tolerance higher than 77% to pH 2.5 compared to their control growth curve at pH 7.0 (Figs. 1a, S1). Regarding bile salts, all strains showed tolerance higher than 80% at a concentration of 0.3% (Figs. 1b, S2).

L. plantarum Strains Inhibit the Growth of Pathogenic Bacteria

All strains could exert an inhibitory effect against pathogenic bacteria, with inhibition zones above 12 mm. Most of them presented potent inhibition (diameter ≥ 20 mm) to all pathogenic bacteria, except for Lpl9, Lpl11, Lpl14, and Lpl19, which exhibited moderate inhibition to *S. Typhimurium* ($20\text{ mm} \leq \text{diameter} \leq 10\text{ mm}$) (Fig. 2).

L. plantarum Strains Adhere to Caco-2 Cells

The strains showed an adhesion percentage ranging from 2.22 to 13.64% (Fig. 3). The strains with higher adherence rates were: Lpl2 (13.64%), Lpl11 (9.05%), Lpl9 (6.51%), Lpl19 (5.72%), and Lpl17 (5.49%), respectively.

Antibiotic Susceptibility, Mucolytic, and Hemolytic Activity of *L. plantarum* Strains

All strains were susceptible to clindamycin, erythromycin, chloramphenicol, and ampicillin antibiotics (Table 3). However, they presented resistance to gentamicin, streptomycin, and vancomycin. Only Lpl4 and Lpl14 strains were susceptible to tetracycline (Table 3). No mucin degradation or hemolysis activity was detected in all nine *L. plantarum* strains tested (data not shown).

Table 2 List of primers used in qPCR

Gene	Forward primer	Reverse primer	Reference
<i>Il10</i>	GGTTGCCAAGCCTTATCGGA	ACCTGCTCCACTGCCTTGCT	[31]
<i>Il6</i>	GAGGATACCACTCCCAACAGACC	AAGTGCATCATCGTTGTTCATACA	[31]
<i>Il1b</i>	CTCCATGAGCTTTGTACAAGG	TGCTGATGTACCAGTTGGGG	[32]
<i>Gapdh</i>	TCACCACCATGGAGAAGGC	GCTAAGCAGTTGGTGGTGCA	[31]

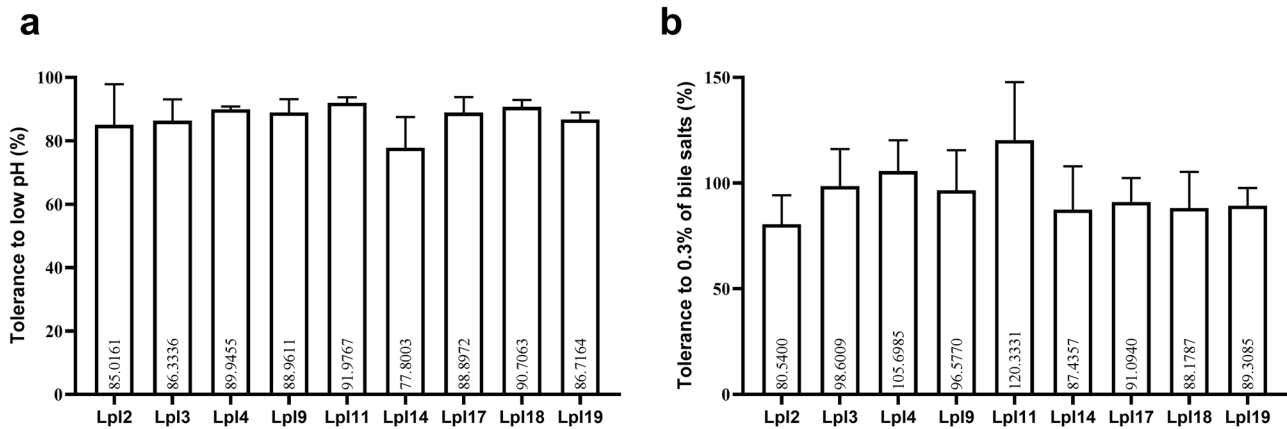


Fig. 1 Tolerance (%) of *L. plantarum* strains to pH 2.5 (a) and 0.3% of bile salts (b). The percentage was obtained by calculating the average difference between the area under the control curve and the test's growth curve

L. plantarum Strains Have Beneficial Effects on 5-FU-Induced Mice's Intestinal Mucositis

Mice Body Weight, Hydric, and Feed Consumption

It was observed a considerable decrease in food (Fig. 4a) and hydric intake (Fig. 4b), and mice body weight (Fig. 4c) after mucositis induced by 5-FU-administration in comparison to the control (CTRL). *L. plantarum* strains administration could not ameliorate these parameters (Fig. 4), except for the Lpl11 group (Fig. 4b), which improved the hydric intake.

L. plantarum Strains Prevent Epithelial Damage in 5-FU-Inflamed Mice

Regarding epithelial damage induced by 5-FU, histopathological alterations such as villus shortening, increased crypt depth, intense inflammatory cell infiltration, and ulceration were observed in the MUC group and correlated with histological scores (Fig. 5a, b). Our results showed that the administration of seven strains (Lpl2, Lpl3, Lpl4, Lpl9, Lpl11, Lpl14, and Lpl18) was able to reduce 5-FU-induced intestinal mucosal damage ($p < 0.05$) (Fig. 5a, b). The same

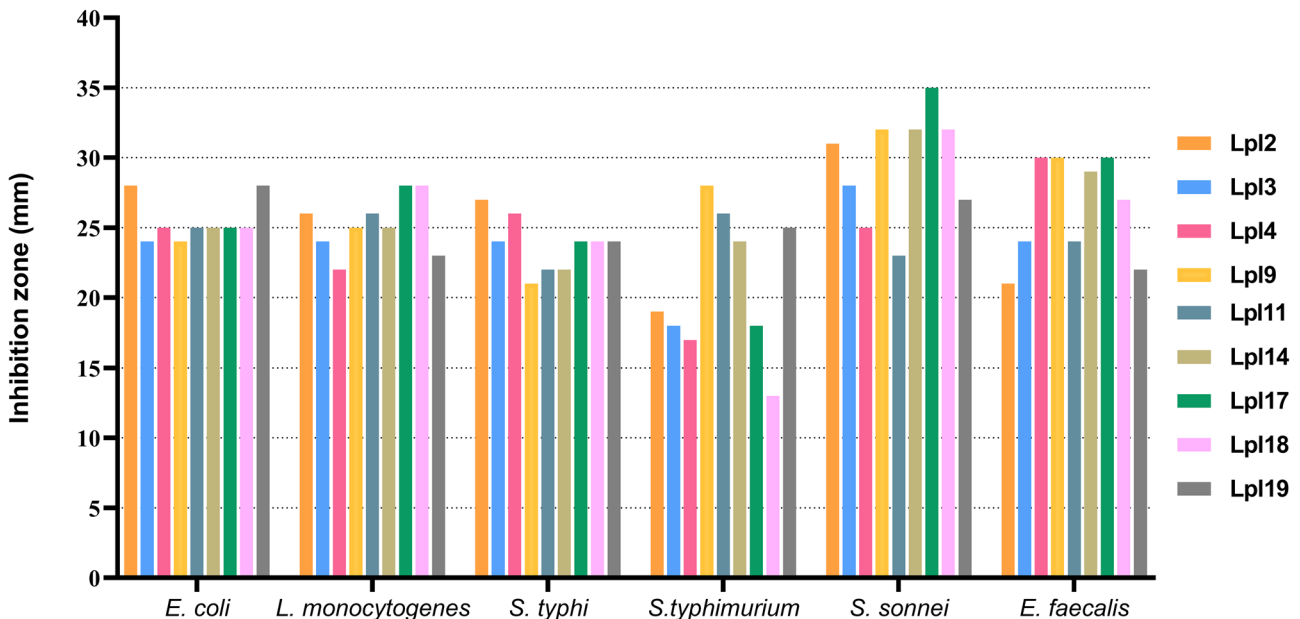


Fig. 2 Antibacterial activity of *L. plantarum* strains against potentially pathogenic bacteria: *Escherichia coli*, *Listeria monocytogenes*, *Salmonella enterica* serovar Typhi, *Salmonella enterica* serovar Typh-

imurium, *Shigella sonnei*, and *Enterococcus faecalis*. Inhibition zones were measured in mm

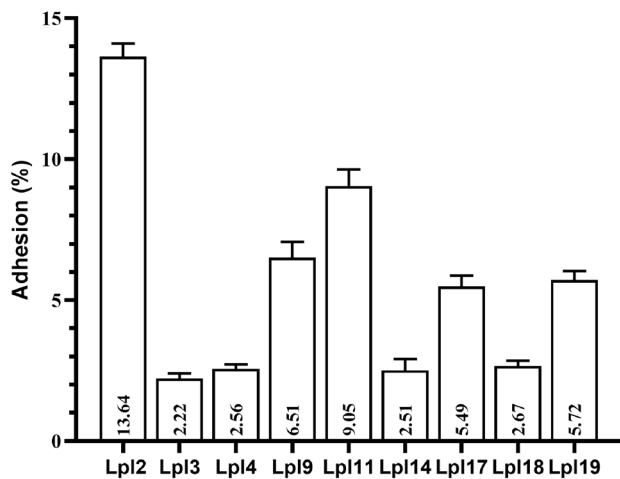


Fig. 3 Percentage of adhesion of *L. plantarum* strains to Caco-2 cells

strains could also significantly prevent villus shortening (Fig. 5c). However, their administration did not promote a considerable effect on crypt depth (Fig. 5d). Even so, the villus height/crypt depth ratio showed an improvement in comparison to the MUC group ($p < 0.05$) (Fig. 5e).

L. plantarum Strains Reduce 5-FU-Induced Intestinal Inflammation

Mice inflamed with 5-FU (MUC group) showed a high level of EPO (1.11 ± 0.07 AU/mg) (Fig. 6a) and MPO (0.40 ± 0.07 AU/mg) (Fig. 6b) activity when compared with the control group (EPO: 0.40 ± 0.14 AU/mg; MPO: 0.19 ± 0.06 AU/mg) ($p < 0.0001$). After *L. plantarum* strains administration it was observed a reduction in neutrophils and eosinophils recruitment in all treated groups ($p < 0.0001$).

On the other hand, mice inflamed with 5-FU (MUC group) displayed a downregulation of *Il10* (0.18 ± 0.06 , $p < 0.01$) (Fig. 6c), and an upregulation in mRNA

expression of *Il6* (6.39 ± 1.56 , $p < 0.0001$) (Fig. 6d) and *Il1b* (4.12 ± 0.66 , $p < 0.001$) (Fig. 6e) when compared to control group (CTRL): *Il6* (1.0 ± 0.26 , $p < 0.001$), *Il1b* (1.00 ± 0.19 , $p < 0.001$), *Il10* (1.00 ± 0.03 , $p < 0.01$) (Fig. 6). The consumption of Lpl4, Lpl9, Lpl11, Lpl14, and Lpl19 strains promoted downregulation in the mRNA gene expression of both *Il6* and *Il1b* (Fig. 6d, e). A downregulation of *Il1b* was also observed in Lpl17 and Lpl18 strains consumption when compared to the inflamed group (MUC) (Fig. 6e). Additionally, the consumption of Lpl2, Lpl9, Lpl14, Lpl17, and Lpl18 upregulated the gene expression of anti-inflammatory cytokine *Il10* (Fig. 6c).

Discussion

Following the guidance from major regulatory agencies, which requires identification, safety, and efficacy assays [1, 39, 40], the present work carried out a series of tests to assess the probiotic potential of nine novel Brazilian *L. plantarum* strains. All strains' identification was confirmed by MALDI-TOF, demonstrating that these strains belong to the *Lactiplantibacillus plantarum* species. After this, functional analysis of these strains was evaluated.

New probiotic candidates must be able to tolerate acidity and bile salts during transit through the GIT [41]. In this work, *L. plantarum* strains showed increased tolerance to acidic pH ($> 80\%$) and bile salts ($> 70\%$). Tolerance to GIT stress has been previously reported in other *L. plantarum* strains. For example, Haghshenas et al. [42] observed that *L. plantarum* 15HN after 3 h of incubation at pH 2.5 presented a survival level of 71 to 76% [42]. A high survival rate for acid stress was also observed in *L. plantarum* GCC_19M1 [43] and *L. plantarum* 15 [44], which presented a survival rate of 93.48–96.97% and 86.40% when submitted to pH 3.0 and 2.0, respectively [43, 44]. Regarding bile salts tolerance, Jacobsen et al. [45], when evaluating the bile tolerance of 47

Table 3 Antimicrobial susceptibility profile of *L. plantarum* strains

Strains	CLI	ERY	CHL	AMP	TET	GEN	STR	VAN
Lpl2	13 S	17 S	19 S	21 S	12 R	0 R	0 R	0 R
Lpl3	18 S	19 S	25 S	17 S	0 R	0 R	0 R	0 R
Lpl4	10 S	19 S	20 S	23 S	14 S	0 R	0 R	0 R
Lpl9	26 S	17 S	20 S	21 S	11 R	0 R	7 R	0 R
Lpl11	26 S	18 S	21 S	24 S	11 R	0 R	0 R	0 R
Lpl14	17 S	18 S	25 S	16 S	14 S	0 R	0 R	0 R
Lpl17	22 S	20 S	16 S	24 S	10 R	0 R	0 R	0 R
Lpl18	11 S	18 S	22 S	20 S	11 R	0 R	0 R	0 R
Lpl19	25 S	14 S	23 S	29 S	10 R	0 R	0 R	0 R

Numbers indicate the halo size (mm). S susceptible, R resistant, CLI clindamycin, ERY erythromycin, CHL chloramphenicol, AMP ampicillin, TET: tetracycline, GEN gentamicin, STR streptomycin, VAN vancomycin

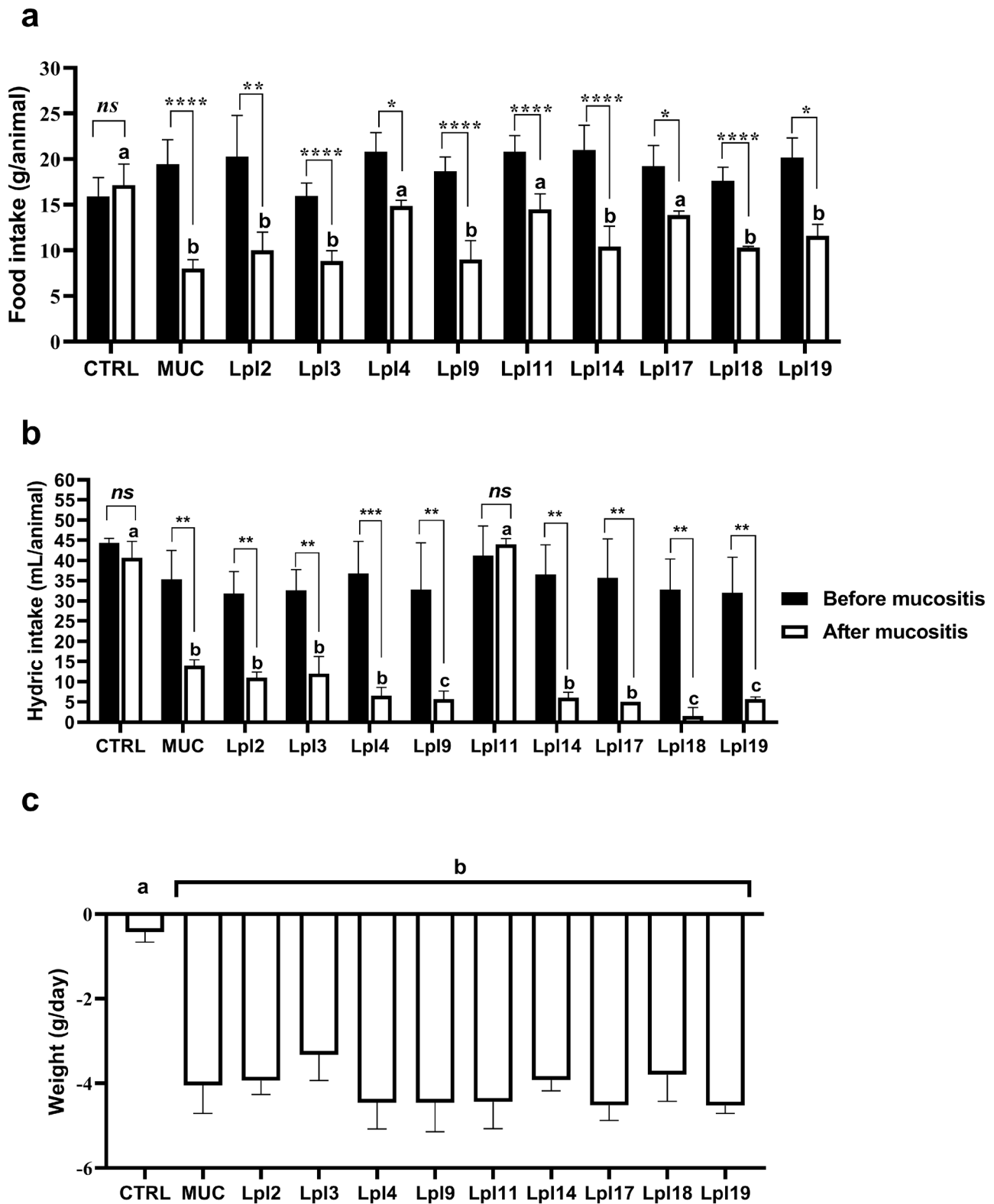


Fig. 4 The effect of *L. plantarum* strains on mice clinical parameters altered by 5-FU. (a) food and (b) liquid intake, and (c) body weight loss. Different letters (a, b) ($p < 0.05$) or asterisks (** $p < 0.01$;

*** $p < 0.001$) indicate statistical differences between the groups. ns indicates no statistical difference

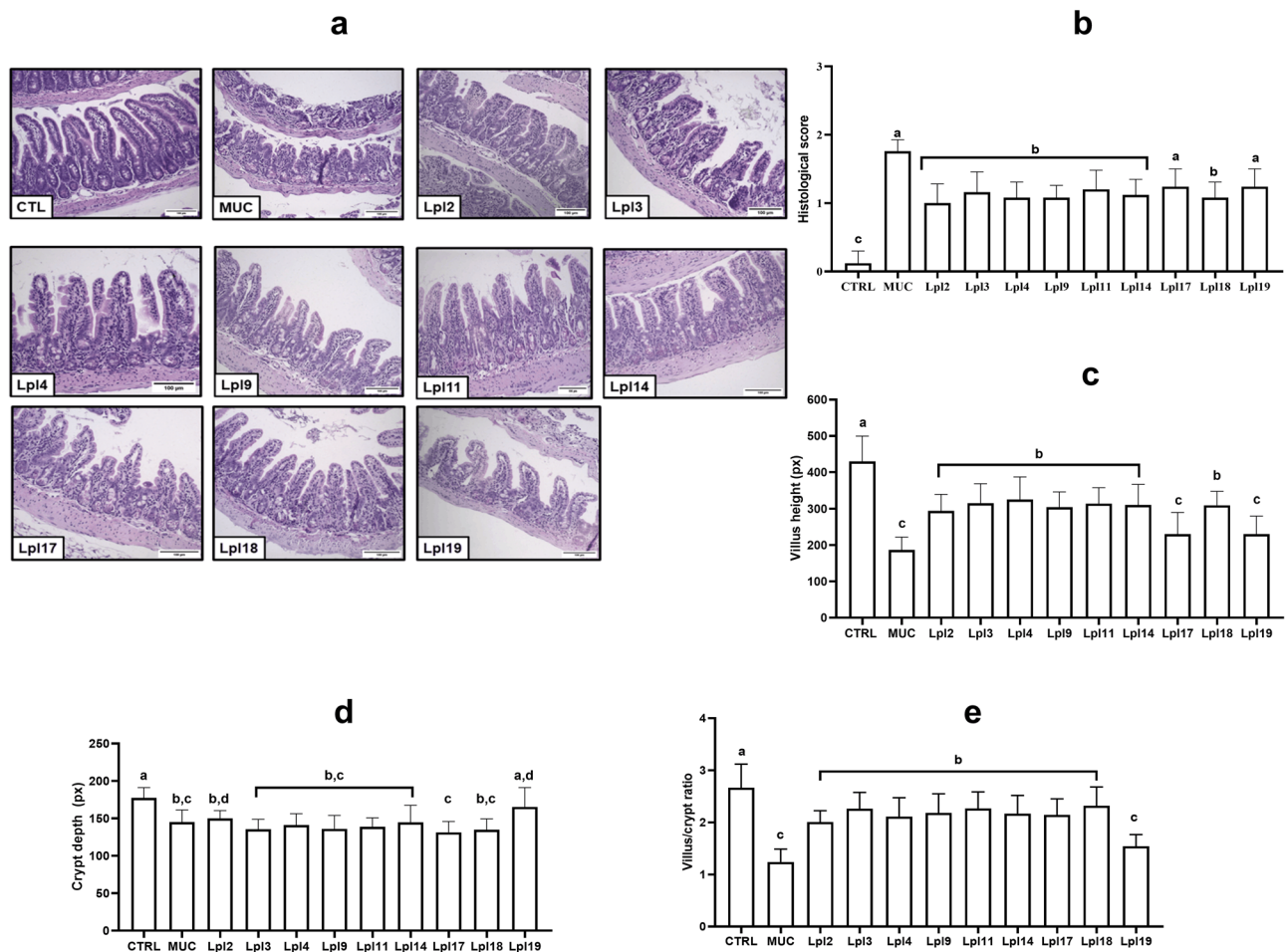


Fig. 5 *L. plantarum* strains prevent histopathological damage induced by 5-FU. (a) Ileum mucosa histopathology, (b) histological score, (c) villus height, (d) crypt depth, and (e) villus height/crypt depth ratio in

pixels (px) (objective: $\times 20$, scale 10 μm). Different letters (a–d) indicate statistical differences between groups ($p < 0.05$)

Lactobacillus strains, including *L. plantarum*, *L. rhamnosus*, *L. acidophilus*, and *L. crispatus*, observed that most of them had relatively high resistance to bile salts (0.3% oxgall) when incubated for 4 h [45]. Similar results were found for *L. plantarum* GCC_19M1, which survived in 0.3% of bile salts [43]. Thus, all these findings evidence that, by tolerating these GIT stressors, the nine *L. plantarum* used in our study can reach the intestine in viable amounts to promote their host-beneficial effects.

After surviving GIT stress, probiotic bacteria candidates should also possess the ability to interact with the intestinal mucosa, which could potentially result in transient colonization, enhancing immunomodulatory effects and inhibiting pathogen growth through the mechanisms of competitive exclusion and bacterial antagonism [46–48]. The present study observed a high adhesion rate to Caco-2 cells by Lpl2, Lp9, Lpl11, Lpl17, and Lpl19. The adherence rates ranged from 0.72–13.64%. These results are supported by Sharma and Kanwar [49], that reported Caco-2 cell

adhesion values ranging from 2.45 to 9.55% when studying lactic acid bacteria isolated from fermented foods. Garcia-Ruiz et al. also reported strain-specific cell adhesion ability [50]. When analyzing adhesion levels to Caco-2 cells of probiotic bacterial strains isolated from wine, these authors obtained adhesion rates ranging from 0.37 to 12.2% [50]. In turn, Behbahani et al. [51] reported adhesion values of 12.2% when analyzing the adhesion of *L. plantarum* L15 to Caco-2 cells [51].

It is interesting to highlight that high adhesion levels of probiotic bacteria to intestinal mucosa can lead to pathogens inhibition, such as *Shigella sonnei* [52], *Listeria monocytogenes* [53], *E. faecalis* pathogenic [54], and *Escherichia coli* [55]. This study observed that Brazilian *L. plantarum* strains have a broad inhibition spectrum against commonly gastrointestinal pathogens. These findings are supported by other studies, which observed that other *L. plantarum* strains could also inhibit potentially pathogenic bacteria, such as *E. coli*, *Staphylococcus aureus*, *E. faecalis*,

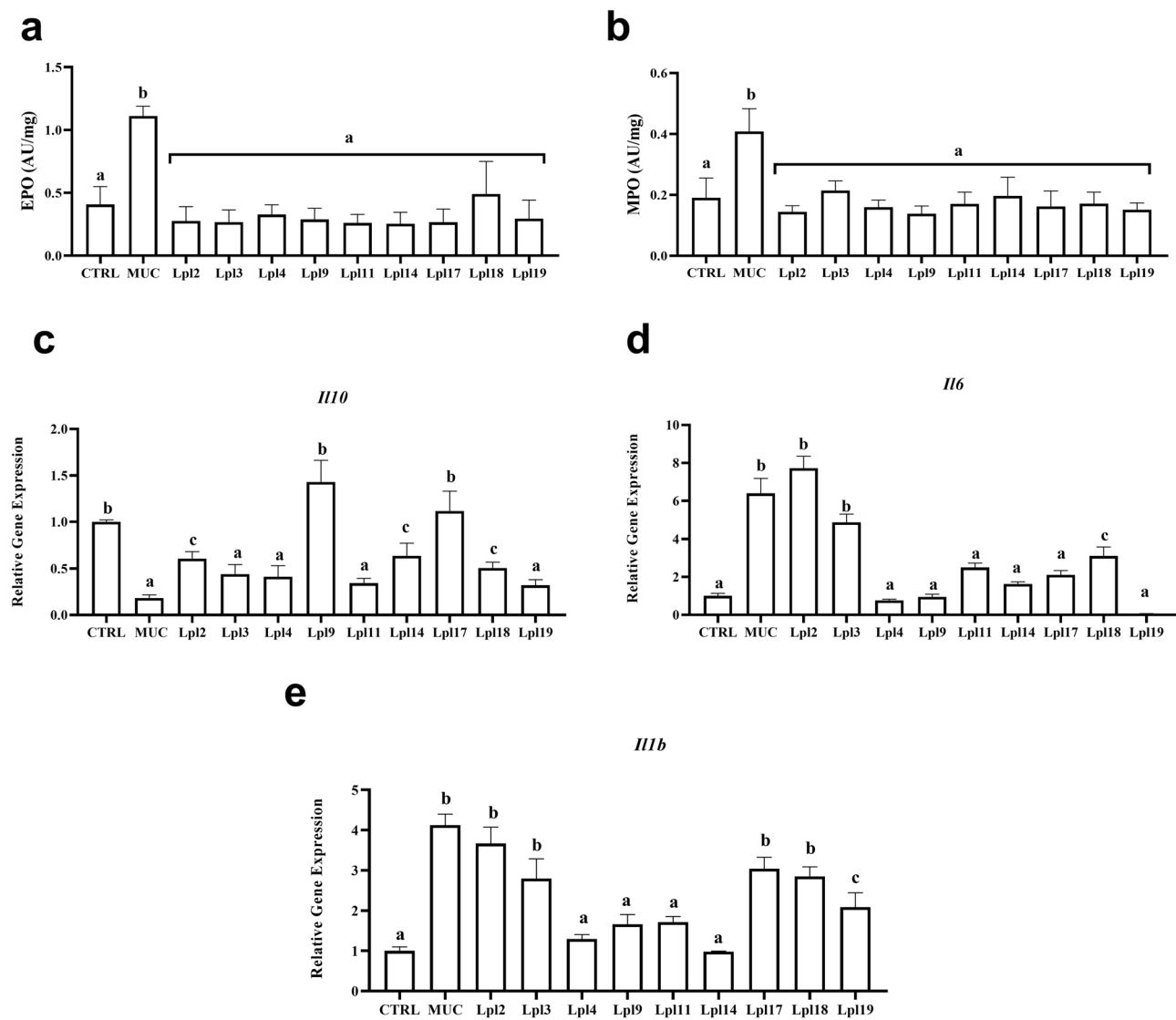


Fig. 6 *L. plantarum* strains ameliorate intestinal inflammation induced by 5-FU. **(a)** eosinophil peroxidase (EPO) and **(b)** myeloperoxidase (MPO) activities; **(c)** *IL10*, **(d)** *IL6*, and **(e)** *IL1b* gene expression. Different letters (a, b, c) indicate statistical differences between groups ($p < 0.05$)

and *L. monocytogenes* [56], *Bacillus cereus*, *Acinetobacter johnsonii*, and *Pseudomonas aeruginosa* strains [43]. This inhibitory property can be related to the production of hydrogen peroxide, organics acid production (e.g., lactic acid or acetate), and bacteriocins (e.g., plantaricin) [57, 58] derived from the fermentation ability of these microorganisms, and also by competing for the binding sites on the intestinal epithelial cells [59]. Due to the high importance of LAB in the daily food industry, the pathogen's inhibition by the *L. plantarum* strains in our study supports their future application as food preservatives or as a potential alternative antimicrobial strategy in the current context of increasing antibiotic resistance among pathogens [60–62].

The potential to adhere to the intestinal mucosa and interact with epithelial cells by probiotic bacteria also

allows them to regulate the activation of the host's immune system. For example, Taverniti et al. [63] demonstrated that *L. helveticus* MIMLh5 reduced the activation of NF- κ B in Caco-2 cells [63]. Other studies showed that pro-inflammatory cytokines *IL1b* and TNF and anti-inflammatory cytokine TGF β were induced by *L. sakei* and *L. johnsonii*, respectively [64].

In our study, the anti-inflammatory effects of the nine *L. plantarum* strains were evaluated through an experimental mice inflammation model of 5-FU (300 mg/kg)-induced intestinal mucositis. This well-established model was chosen due to its severity, which would make it possible to observe more significant protective effects promoted by potential probiotic strains [21, 65]. Additionally, it is essential to highlight that studies evaluating the effect of

L. plantarum in the mucositis model are lacking, with the main findings provided by Levit et al. [27] with *L. plantarum* CLR2130 and Ciobanu et al. [28] with *L. plantarum* ATCC 8014.

The data presented in our study are consistent with the pathobiology of mucositis induced by chemotherapy [21, 36, 65], in which it was observed mice body weight loss, epithelial architecture alteration, villus shortening, and polymorphonuclear cell infiltration in the intestinal mucosa after administration of 5-FU (300 mg/kg). Consumption of all nine Brazilian *L. plantarum* strains reduced eosinophils peroxidase and myeloperoxidase activity, demonstrating their beneficial effect on protecting the intestinal mucosa from polymorphonuclear cells infiltration. The ameliorative effect of these strains on these parameters can reduce the oxidative stress and production of pro-inflammatory cytokines induced by these cells, thus preventing the exacerbation of tissue damage and mucositis progression [17]. Similar findings were also previously observed in other probiotics, such as *Lactocaseibacillus rhamnosus* [66], *Rhodotorula mucilaginosa* [67], *L. delbrueckii* [68], *Bifidobacterium longum* [69], among others.

Inflammatory factors play a central role in mucositis's release in the inflammatory phase [70]. Another essential mechanism these strains use to mitigate intestinal inflammation induced by 5-FU includes the modulation of inflammatory and immunoregulatory cytokines. Our results showed that administration of some *L. plantarum* (Lp14, Lp19, Lp111, Lp114, Lp117, Lp118, and Lp119) promoted downregulation of *Il6* and *Il1b* pro-inflammatory cytokines. On the other hand, it was also observed that other strains (Lp12, Lp19, Lp14, Lp17, and Lp18) could upregulate gene expression of the immunoregulatory cytokine *Il10*, a potential anti-inflammatory cytokine related to inflammation resolution during intestinal inflammatory processes [71, 72]. These findings are corroborated by previous works that showed that administration of probiotics such as *L. acidophilus* [73, 74], *L. casei* [18], *Bifidobacterium* sp. [20], and *L. plantarum* strains [27, 28] was able to ameliorate intestinal damage and inflammation induced by 5-FU through modulation of inflammatory cytokines *Il6*, *Ifn γ* , *Il1b*, *Tnf*, and *Il10*.

Bacteria cell components (e.g., surface layer proteins, peptidoglycans, teichoic, and lipoteichoic acids, and exopolysaccharide) or bioactive compounds (e.g., secreted proteins, SCFA) produced by these microorganisms have been identified as the molecular effectors of some health-promoting attributes mediated by probiotics [75–78], due interacting with epithelial cell receptors (including TLRs, GPR43) and inhibiting inflammatory signaling pathways (e.g., NF- κ B and MAPK) [79]. Thus, we suggest that these genetic factors can be involved with the immunomodulatory

effects of the *L. plantarum* strains of our study against 5-FU-induced epithelial inflammation. However, more studies should be conducted to elucidate better the action mechanisms of these strains associated with their intestinal anti-inflammatory effects.

Knowing that the modulation of pro-and anti-inflammatory cytokines is fundamental to controlling the inflammatory damage exacerbation in 5-FU-induced mucositis [17], upregulation of *Il10* cytokine in our study further highlights the anti-inflammatory effect that Brazilian *L. plantarum* strains can exert on the host, ratifying their beneficial effects and use as probiotics. This anti-inflammatory property can be associated with the ability of these strains to improve the intestinal epithelium architecture observed through reduction of villous atrophy, restoration of the villus height/crypt depth ratio, and improvement of histopathological score (Lp12, Lp13, Lp14, Lp11, Lp19, and Lp18). Improving these parameters can lead to a better clinical status of the mice, including body weight.

In this context, although the single strains have a positive impact in ameliorating intestinal inflammation induced by 5-FU, none of the tested strains was able to prevent weight loss. These results are supported by other studies which observed improvements in some mucositis parameters after probiotics administration regardless of improvement in water/feed consumption or body weight [18, 20, 37]. Based on this information, some studies have demonstrated that the association of two or more probiotic strains can be a strategy to enhance their beneficial effects, including ameliorating the mice's body weight [80, 81]. Therefore, the strains characterized in this study can also be associated to maximize their effects, opening a wide range of studies and exploring possibilities.

It is also interesting to highlight that safety issues must be evaluated during the characterization of new probiotic candidates. Thus, this study evaluated the antibiotic resistance profile and mucin and hemolytic activity of *L. plantarum* strains. None of them presented hemolytic or mucolytic activity. These findings are supported by other studies that demonstrated no hemolytic or mucolytic activity by LAB potential probiotic strains such as *L. delbrueckii* CIDCA 133 [82], *L. plantarum* BCC9546 [83], and *L. helveticus* D75 [84]. Mucin degradation and hemolysis process are infection mechanisms used by pathogenic bacteria. Thus, the lack of this phenotype in *L. plantarum* strains is an indication that these strains do not possess the ability to act as a pathogen and, therefore, would not present a potential risk for host epithelial barrier or blood cell disruption [34, 47, 84]. However, we believe that further in vivo safety studies of these strains should be performed to support our in vitro findings.

Antibiotic resistance is another main concern regarding probiotics' safety. In our study, some *L. plantarum* strains

presented a resistance profile to gentamicin, streptomycin, and vancomycin, corroborating with other studies. For example, Hummel et al. [85] found that among forty-five LAB strains, aminoglycosides (gentamicin and streptomycin) and ciprofloxacin resistance were prevalent in more than 70% of them, indicating that maybe this could be an intrinsic resistance mechanism [85]. The same study observed a low resistance profile for ampicillin, penicillin, chloramphenicol, and tetracycline [85]. Anisimova and Yarullina [86] analyzed the antibiotic resistance profile of twenty *Lactobacillus* strains and showed that 90% were sensible to clindamycin and 95% to tetracycline, respectively erythromycin, and rifampicin. All of them were susceptible to chloramphenicol and ampicillin. As expected, most were resistant to vancomycin and aminoglycosides [86].

Lactobacillus species are known to be intrinsically resistant to vancomycin and aminoglycosides [87], as observed in all strains in this work, but this constitutive resistance is not considered a matter of concern [86, 88, 89]. Based on the phenotypic findings of antimicrobial resistance, further studies should be conducted to evaluate the resistance nature observed in the *L. plantarum* strains described in our study, once the antibiotic resistance becomes a problem when resistance is transferable by mobile elements such as plasmids [39, 88, 89]. Thus, in addition to phenotypic profiling, genomic analysis should also be applied to identify whether genes associated with antimicrobial resistance are present in the chromosome or mobile elements of these microorganisms [90].

Conclusions

In conclusion, this study revealed the strain-dependent functionality and hosted beneficial effects of *L. plantarum* strains isolated from different sources in Brazil. Most strains tested in vitro present good safe aspects and probiotic features to be explored in different fields, such as in the food industry, biotechnology, nutrition, and biotherapeutic approaches. Additionally, most of them were able, at some points, to alleviate intestinal inflammation induced by 5-FU chemotherapy in the mouse model, evidencing the anti-inflammatory and protective effects on an intestinal level. The characterization of the potential probiotic property of these strains opens perspectives for new studies to be carried out to evaluate their biotherapeutic property in other inflammatory conditions, which help to fully explain the mechanisms used for these strains to promote their beneficial effects.

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Data Availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics Approval The study was approved by the Local Animal Experimental Ethics Committee of the Federal University of Minas Gerais (CEUA-UFMG) (Protocol number. 112/2020) and conducted according to the guidelines of the Brazilian College of Animal Experimentation (COBEA).

Conflict of Interest The authors declare no competing interests.

References

- Hill C, Guarner F, Reid G et al (2014) The International Scientific Association for Probiotics and Prebiotics consensus statement on the scope and appropriate use of the term probiotic. *Nat Rev Gastroenterol Hepatol* 11:506–514. <https://doi.org/10.1038/nrgastro.2014.66>
- Markowiak P, Śliżewska K (2017) Effects of probiotics, prebiotics, and synbiotics on human health. *Nutrients* 9:1021. <https://doi.org/10.3390/nu9091021>
- Klein G, Pack A, Bonaparte C, Reuter G (1998) Taxonomy and physiology of probiotic lactic acid bacteria. *Int J Food Microbiol* 41:103–125. [https://doi.org/10.1016/S0168-1605\(98\)00049-X](https://doi.org/10.1016/S0168-1605(98)00049-X)
- Zheng J, Wittouck S, Salvetti E et al (2020) A taxonomic note on the genus *Lactobacillus*: Description of 23 novel genera, emended description of the genus *Lactobacillus* Beijerinck 1901, and union of Lactobacillaceae and Leuconostocaceae. *Int J Syst Evol Microbiol* 70:2782–2858. <https://doi.org/10.1099/ijsem.0.004107>
- Crittenden RG, Martinez NR, Playne MJ (2003) Synthesis and utilisation of folate by yoghurt starter cultures and probiotic bacteria. *Int J Food Microbiol* 80:217–222. [https://doi.org/10.1016/S0168-1605\(02\)00170-8](https://doi.org/10.1016/S0168-1605(02)00170-8)
- Saier MH Jr, Mansour NM (2005) Probiotics and prebiotics in Human Health. *J Mol Microbiol Biotechnol* 10:22–25. <https://doi.org/10.1159/000090345>
- Liu Y-W, Liong M-T, Tsai Y-C (2018) New perspectives of *Lactobacillus plantarum* as a probiotic: The gut-heart-brain axis. *J Microbiol* 56:601–613. <https://doi.org/10.1007/s12275-018-8079-2>
- Li S, Huang R, Shah NP et al (2014) Antioxidant and antibacterial activities of exopolysaccharides from *Bifidobacterium bifidum*

- WBIN03 and *Lactobacillus plantarum* R315. J Dairy Sci 97:7334–7343. <https://doi.org/10.3168/jds.2014-7912>
9. Hu C, Ren L, Zhou Y, Ye B (2019) Characterization of antimicrobial activity of three *Lactobacillus plantarum* strains isolated from Chinese traditional dairy food. Food Sci Nutr 7:1997–2005. <https://doi.org/10.1002/fsn3.1025>
 10. Park S, Ji Y, Jung H-Y et al (2017) *Lactobacillus plantarum* HAC01 regulates gut microbiota and adipose tissue accumulation in a diet-induced obesity murine model. Appl Microbiol Biotechnol 101:1605–1614. <https://doi.org/10.1007/s00253-016-7953-2>
 11. Lim S-D, Kim S, Huang E et al (2018) Physiological characteristics and anti-obesity effect of *Lactobacillus plantarum* K10. Korean J Food Sci An 38:2234–2246. <https://doi.org/10.5851/kosfa.2018.38.3.554>
 12. Fuentes MC, Lajo T, Carrión JM, Cuñé J (2013) Cholesterol-lowering efficacy of *Lactobacillus plantarum* CECT 7527, 7528 and 7529 in hypercholesterolaemic adults. Br J Nutr 109:1866–1872. <https://doi.org/10.1017/S000711451200373X>
 13. Le B, Yang SH (2018) Efficacy of *Lactobacillus plantarum* in prevention of inflammatory bowel disease. Toxicol Reports 5:314–317. <https://doi.org/10.1016/j.toxrep.2018.02.007>
 14. Zhang F, Li Y, Wang X et al (2019) The Impact of *Lactobacillus plantarum* on the Gut microbiota of mice with DSS-induced colitis. Biomed Res Int 2019:1–10. <https://doi.org/10.1155/2019/3921315>
 15. Yue Y, Ye K, Lu J et al (2020) Probiotic strain *Lactobacillus plantarum* YYC-3 prevents colon cancer in mice by regulating the tumour microenvironment. Biomed Pharmacother 127:110159. <https://doi.org/10.1016/j.biopha.2020.110159>
 16. Tallon R, Arias S, Bressollier P, Urdaci MC (2007) Strain- and matrix-dependent adhesion of *Lactobacillus plantarum* is mediated by proteinaceous bacterial compounds. J Appl Microbiol 102:442–451. <https://doi.org/10.1111/j.1365-2672.2006.03086.x>
 17. Batista VL, da Silva TF, de Jesus LCL et al (2020) Probiotics, prebiotics, synbiotics, and paraprobiotics as a therapeutic alternative for intestinal mucositis. Front Microbiol 11:544490. <https://doi.org/10.3389/fmicb.2020.544490>
 18. Yeung C-Y, Chan W-T, Jiang C-B et al (2015) Amelioration of chemotherapy-induced intestinal mucositis by orally administered probiotics in a mouse model. PLoS ONE 10:e0138746. <https://doi.org/10.1371/journal.pone.0138746>
 19. Tang Y, Wu Y, Huang Z et al (2017) Administration of probiotic mixture DM#1 ameliorated 5-fluorouracil-induced intestinal mucositis and dysbiosis in rats. Nutrition 33:96–104. <https://doi.org/10.1016/j.nut.2016.05.003>
 20. Quaresma M, Damasceno S, Monteiro C et al (2020) Probiotic mixture containing *Lactobacillus* spp. and *Bifidobacterium* spp. attenuates 5-fluorouracil-induced intestinal mucositis in mice. Nutr Cancer 72:1355–1365. <https://doi.org/10.1080/01635581.2019.1675719>
 21. Sougiannis AT, VanderVeen BN, Davis JM et al (2021) Understanding chemotherapy-induced intestinal mucositis and strategies to improve gut resilience. Am J Physiol Liver Physiol 320:G712–G719. <https://doi.org/10.1152/ajpgi.00380.2020>
 22. Dahlgren D, Sjöblom M, Hellström PM, Lennernäs H (2021) Chemotherapeutics-Induced intestinal mucositis: pathophysiology and potential treatment strategies. Front Pharmacol 12:681417. <https://doi.org/10.3389/fphar.2021.681417>
 23. Cinausero M, Aprile G, Ermacora P et al (2017) New frontiers in the pathobiology and treatment of cancer regimen-related mucosal injury. Front Pharmacol 8:354. <https://doi.org/10.3389/fphar.2017.00354>
 24. Stringer AM (2013) Interaction between host cells and microbes in chemotherapy-induced mucositis. Nutrients 5:1488–1499. <https://doi.org/10.3390/nu5051488>
 25. Touchefeu Y, Montassier E, Nieman K et al (2014) Systematic review: the role of the gut microbiota in chemotherapy- or radiation-induced gastrointestinal mucositis - current evidence and potential clinical applications. Aliment Pharmacol Ther 40:409–421. <https://doi.org/10.1111/apt.12878>
 26. Gori S, Inno A, Belluomini L et al (2019) Gut microbiota and cancer: How gut microbiota modulates activity, efficacy and toxicity of antitumoral therapy. Crit Rev Oncol Hematol 143:139–147. <https://doi.org/10.1016/j.critrevonc.2019.09.003>
 27. Levit R, Savoy de Giori G, de Moreno de LeBlanc A, LeBlanc JG (2018) Protective effect of the riboflavin-overproducing strain *Lactobacillus plantarum* CRL2130 on intestinal mucositis in mice. Nutrition 54:165–172. <https://doi.org/10.1016/j.nut.2018.03.056>
 28. Ciobanu L, Tefas C, Oancea D et al (2020) Effect of *Lactobacillus plantarum* ACTT 8014 on 5-fluorouracil induced intestinal mucositis in Wistar rats. Exp Ther Med 20:1–1. <https://doi.org/10.3892/etm.2020.9339>
 29. Tokatlı M, Gülgör G, Bağder Elmacı S et al (2015) In vitro properties of potential probiotic indigenous lactic acid bacteria originating from traditional pickles. Biomed Res Int 2015:1–8. <https://doi.org/10.1155/2015/315819>
 30. Walker DK, Gilliland SE (1993) Relationships among bile tolerance, bile salt deconjugation, and assimilation of cholesterol by *Lactobacillus acidophilus*. J Dairy Sci 76:956–961. [https://doi.org/10.3168/jds.S0022-0302\(93\)77422-6](https://doi.org/10.3168/jds.S0022-0302(93)77422-6)
 31. Toure R, Kheadr E, Lacroix C et al (2003) Production of antibacterial substances by bifidobacterial isolates from infant stool active against *Listeria monocytogenes*. J Appl Microbiol 95:1058–1069. <https://doi.org/10.1046/j.1365-2672.2003.02085.x>
 32. Nami Y, Abdullah N, Haghsheenas B et al (2014) Probiotic assessment of *Enterococcus durans* 6HL and *Lactococcus lactis* 2HL isolated from vaginal microflora. J Med Microbiol 63:1044–1051. <https://doi.org/10.1099/jmm.0.074161-0>
 33. Clinical and Laboratory Standards Institute (CLSI) (2018) Performance standards for antimicrobial susceptibility testing. 28th ed. CLSI supplement M100. Wayne, PA. Diponible in: <https://file.qums.ac.ir/repository/mmr/CLSI-2018-M100-S28.pdf>
 34. Zhou JS, Gopal PK, Gill HS (2001) Potential probiotic lactic acid bacteria *Lactobacillus rhamnosus* (HN001), *Lactobacillus acidophilus* (HN017) and *Bifidobacterium lactis* (HN019) do not degrade gastric mucin in vitro. Int J Food Microbiol 63:81–90. [https://doi.org/10.1016/S0168-1605\(00\)00398-6](https://doi.org/10.1016/S0168-1605(00)00398-6)
 35. Xu L, Yuan J, Chen X et al (2021) Screening of intestinal probiotics and the effects of feeding probiotics on the digestive enzyme activity, immune, intestinal flora and WSSV resistance of *Procambarus clarkii*. Aquaculture 540:736748. <https://doi.org/10.1016/j.aquaculture.2021.736748>
 36. Carvalho RD, Breyner N, Menezes-Garcia Z et al (2017) Secretion of biologically active pancreatitis-associated protein I (PAP) by genetically modified dairy *Lactococcus lactis* NZ9000 in the prevention of intestinal mucositis. Microb Cell Fact 16:27. <https://doi.org/10.1186/s12934-017-0624-x>
 37. Soares PMG, Mota JMSc, Gomes AS et al (2008) Gastrointestinal dysmotility in 5-fluorouracil-induced intestinal mucositis outlasts inflammatory process resolution. Cancer Chemother Pharmacol 63:91–98. <https://doi.org/10.1007/s00280-008-0715-9>
 38. De Jesus LCL, Drumond MM, de Carvalho A et al (2019) Protective effect of *Lactobacillus delbrueckii* subsp. *lactis* CIDCA 133 in a model of 5 Fluorouracil-Induced intestinal mucositis. J Funct Foods 53:197–207. <https://doi.org/10.1016/j.jff.2018.12.027>
 39. Rychen G, Aquilina G, Azimonti G et al (2018) Guidance on the characterisation of microorganisms used as feed additives or as production organisms. EFSA J. <https://doi.org/10.2903/j.efsa.2018.5206>
 40. Anvisa (2021) Guia de Instrução Processual de Petição de Avaliação de Probióticos para Uso em Alimentos. In: Guid. no

- 21/2021 – version 2. Disponible in: <http://antigo.anvisa.gov.br/documents/10181/5280930/guia+21+v2.pdf/dac5bf5f-ae56-4444-b53c-2cf0f7c15301>
41. Papadimitriou K, Alegría Á, Bron PA et al (2016) Stress physiology of lactic acid bacteria. *Microbiol Mol Biol Rev* 80:837–890. <https://doi.org/10.1128/MMBR.00076-15>
 42. Haghshenas B, Nami Y, Almasi A et al (2017) Isolation and characterization of probiotics from dairies. *Iran J Microbiol* 9:234–243
 43. Nath S, Sikidar J, Roy M, Deb B (2020) In vitro screening of probiotic properties of *Lactobacillus plantarum* isolated from fermented milk product Graphical Abstract. *Food Qual Saf* 4:213–223. <https://doi.org/10.1093/fqsafe/fyaa026>
 44. Riane K (2019) Probiotic properties and antioxidant efficiency of *Lactobacillus plantarum* 15 isolated from milk. *J Microbiol Biotechnol Food Sci* 9:516–520. <https://doi.org/10.15414/jmbfs.2019/20.9.3.516-520>
 45. Jacobsen CN, Rosenfeldt Nielsen V, Hayford AE et al (1999) Screening of probiotic activities of forty-seven strains of *Lactobacillus* spp. by In vitro techniques and evaluation of the colonization ability of five selected strains in humans. *Appl Environ Microbiol* 65:4949–4956. <https://doi.org/10.1128/AEM.65.11.4949-4956.1999>
 46. Servin AL, Coconnier M-H (2003) Adhesion of probiotic strains to the intestinal mucosa and interaction with pathogens. *Best Pract Res Clin Gastroenterol* 17:741–754. [https://doi.org/10.1016/S1521-6918\(03\)00052-0](https://doi.org/10.1016/S1521-6918(03)00052-0)
 47. Monteagudo-Mera A, Rastall RA, Gibson GR et al (2019) Adhesion mechanisms mediated by probiotics and prebiotics and their potential impact on human health. *Appl Microbiol Biotechnol* 103:6463–6472. <https://doi.org/10.1007/s00253-019-09978-7>
 48. Velez MP, De Keersmaecker SCJ, Vanderleyden J (2007) Adherence factors of *Lactobacillus* in the human gastrointestinal tract. *FEMS Microbiol Lett* 276:140–148. <https://doi.org/10.1111/j.1574-6968.2007.00908.x>
 49. Sharma S, Kanwar SS (2017) Adherence potential of indigenous lactic acid bacterial isolates obtained from fermented foods of Western Himalayas to intestinal epithelial Caco-2 and HT-29 cell lines. *J Food Sci Technol* 54:3504–3511. <https://doi.org/10.1007/s13197-017-2807-1>
 50. García-Ruiz A, González de Llano D, Esteban-Fernández A et al (2014) Assessment of probiotic properties in lactic acid bacteria isolated from wine. *Food Microbiol* 44:220–225. <https://doi.org/10.1016/j.fm.2014.06.015>
 51. Alizadeh Behbahani B, Noshad M, Falah F (2019) Inhibition of *Escherichia coli* adhesion to human intestinal Caco-2 cells by probiotic candidate *Lactobacillus plantarum* strain L15. *Microb Pathog* 136:103677. <https://doi.org/10.1016/j.micpath.2019.103677>
 52. Zhang Y-C, Zhang L-W, Ma W et al (2012) Screening of probiotic lactobacilli for inhibition of *Shigella sonnei* and the macromolecules involved in inhibition. *Anaerobe* 18:498–503. <https://doi.org/10.1016/j.anaerobe.2012.08.007>
 53. de Castilho NPA, Todorov SD, Oliveira LL et al (2020) Inhibition of *Listeria monocytogenes* in fresh sausage by bacteriocinogenic *Lactobacillus curvatus* UFV-NPAC1 and its semi-purified bacteriocin. *LWT* 118:108757. <https://doi.org/10.1016/j.lwt.2019.108757>
 54. Bohora AA, Kokate SR (2017) Good Bugs vs Bad Bugs: evaluation of inhibitory effect of selected probiotics against *Enterococcus faecalis*. *J Contemp Dent Pract* 18:312–316. <https://doi.org/10.5005/jp-journals-10024-2037>
 55. Halder D, Mandal M, Chatterjee S et al (2017) Indigenous probiotic *Lactobacillus* isolates presenting antibiotic like activity against human pathogenic bacteria. *Biomedicines* 5:31. <https://doi.org/10.3390/biomedicines5020031>
 56. Zangeneh M, Khorrami S, Khaleghi M (2020) Bacteriostatic activity and partial characterization of the bacteriocin produced by *L. plantarum* sp. isolated from traditional sourdough. *Food Sci Nutr* 8:6023–6030. <https://doi.org/10.1002/fsn3.1890>
 57. de Sousa MA, Rama GR, Volken de Souza CF, Granada CE (2020) Acid lactic lactobacilli as a biotechnological toll to improve food quality and human health. *Biotechnol Prog*. <https://doi.org/10.1002/btpr.2937>
 58. Cazorla SI, Maldonado-Galdeano C, Weill R et al (2018) Oral Administration of probiotics increases paneth cells and intestinal antimicrobial activity. *Front Microbiol* 9:736. <https://doi.org/10.3389/fmicb.2018.00736>
 59. Campana R, van Hemert S, Baffone W (2017) Strain-specific probiotic properties of lactic acid bacteria and their interference with human intestinal pathogens invasion. *Gut Pathog* 9:12. <https://doi.org/10.1186/s13099-017-0162-4>
 60. Hernández-González JC, Martínez-Tapia A, Lazcano-Hernández G et al (2021) Bacteriocins from Lactic Acid Bacteria. A powerful alternative as antimicrobials, probiotics, and immunomodulators in veterinary medicine. *Animals* 11:979. <https://doi.org/10.3390/ani11040979>
 61. Ibrahim SA, Ayivi RD, Zimmerman T et al (2021) Lactic acid bacteria as antimicrobial agents: food safety and microbial food spoilage prevention. *Foods* 10:3131. <https://doi.org/10.3390/foods10123131>
 62. Oldak A, Zielińska D (2017) Bacteriocins from lactic acid bacteria as an alternative to antibiotics. *Postepy Hig Med Dosw*. <https://doi.org/10.5604/01.3001.0010.3817>
 63. Taverniti V, Stuknyte M, Minuzzo M et al (2013) S-layer protein mediates the stimulatory effect of *Lactobacillus helveticus* MIMLh5 on innate immunity. *Appl Environ Microbiol* 79:1221–1231. <https://doi.org/10.1128/AEM.03056-12>
 64. Haller D, Bode C, Hammes WP et al (2000) Non-pathogenic bacteria elicit a differential cytokine response by intestinal epithelial cell/leucocyte co-cultures. *Gut* 47:79–87. <https://doi.org/10.1136/gut.47.1.79>
 65. Sonis ST, Tracey C, Shklar G et al (1990) An animal model for mucositis induced by cancer chemotherapy. *Oral Surgery, Oral Med Oral Pathol* 69:437–443. [https://doi.org/10.1016/0030-4220\(90\)90376-4](https://doi.org/10.1016/0030-4220(90)90376-4)
 66. Trindade LM, Torres L, Matos ID et al (2021) Paraprobiotic *Lactocaseibacillus rhamnosus* Protects intestinal damage in an experimental murine model of mucositis. *Probiotics Antimicrob Proteins*. <https://doi.org/10.1007/s12602-021-09842-z>
 67. Coutinho JOPA, Quintanilha MF, Campos MRA et al (2021) Antarctic strain of *Rhodotorula mucilaginosa* UFMGCB 18,377 Attenuates mucositis induced by 5-fluorouracil in mice. *Probiotics Antimicrob Proteins*. <https://doi.org/10.1007/s12602-021-09817-0>
 68. Barroso FAL, de Jesus LCL, da Silva TF et al (2022) *Lactobacillus delbrueckii* CIDCA 133 Ameliorates chemotherapy-induced mucositis by modulating epithelial barrier and TLR2/4/Myd88/NF- κ B signaling pathway. *Front Microbiol*. <https://doi.org/10.3389/fmicb.2022.858036>
 69. Quintanilha MF, Miranda VC, Souza RO et al (2022) *Bifidobacterium longum* subsp. *longum* 51A attenuates intestinal injury against irinotecan-induced mucositis in mice. *Life Sci* 289:120243. <https://doi.org/10.1016/j.lfs.2021.120243>
 70. Sonis ST (2004) The pathobiology of mucositis. *Nat Rev Cancer* 4:277–284. <https://doi.org/10.1038/nrc1318>
 71. Denning TL, Campbell NA, Song F et al (2000) Expression of IL-10 receptors on epithelial cells from the murine small and large intestine. *Int Immunol* 12:133–139. <https://doi.org/10.1093/intimm/12.2.133>
 72. Jarry A, Bossard C, Bou-Hanna C et al (2008) Mucosal IL-10 and TGF- β play crucial roles in preventing LPS-driven, IFN- γ -mediated epithelial damage in human colon explants. *J Clin Invest*. <https://doi.org/10.1172/JCI32140>

73. Justino PF, Melo LF, Nogueira AF, Morais C et al (2015) Regulatory role of *Lactobacillus acidophilus* on inflammation and gastric dysmotility in intestinal mucositis induced by 5-fluorouracil in mice. *Cancer Chemother Pharmacol* 3:559–567. <https://doi.org/10.1007/s00280-014-2663-x>
74. Oh NS, Lee JY, Lee JM et al (2017) Mulberry leaf extract fermented with *Lactobacillus acidophilus* A4 ameliorates 5-fluorouracil-induced intestinal mucositis in rats. *Lett Appl Microbiol* 64:459–468. <https://doi.org/10.1111/lam.12741>
75. Sengupta R, Altermann E, Anderson RC et al (2013) The Role of cell surface architecture of Lactobacilli in host-microbe interactions in the gastrointestinal tract. *Mediators Inflamm* 2013:1–16. <https://doi.org/10.1155/2013/237921>
76. Chandhni PR, Pradhan D, Sowmya K et al (2021) Ameliorative effect of surface proteins of probiotic lactobacilli in colitis mouse models. *Front Microbiol*. <https://doi.org/10.3389/fmicb.2021.679773>
77. De Jesus LCL, Aburjaile FF, Sousa TDJ et al (2022) Genomic characterization of *Lactobacillus delbrueckii* strains with probiotics properties. *Front Bioinforma*. <https://doi.org/10.3389/fbinf.2022.912795>
78. Ruiz L, Hevia A, Bernardo D et al (2014) Extracellular molecular effectors mediating probiotic attributes. *FEMS Microbiol Lett* 359:1–11. <https://doi.org/10.1111/1574-6968.12576>
79. Delgado S, Sánchez B, Margolles A et al (2020) Molecules produced by probiotics and intestinal microorganisms with immunomodulatory activity. *Nutrients* 12:391. <https://doi.org/10.3390/nu12020391>
80. Gupta N, Ferreira J, Hsu C et al (2020) *Lactobacillus reuteri* DSM 17938 and ATCC PTA 5289 ameliorates chemotherapy-induced oral mucositis. *Sci Reports* 10:16189. <https://doi.org/10.1038/s41598-020-73292-w>
81. Bowen JM, Stringer AM, Gibson RJ et al (2007) VSL#3 probiotic treatment reduces chemotherapy-induced diarrhoea and weight loss. *Cancer Biol Ther* 6:1445–1450. <https://doi.org/10.4161/cbt.6.9.4622>
82. de Jesus LCL, de Jesus Sousa T, Coelho-Rocha ND et al (2021) Safety evaluation of *Lactobacillus delbrueckii* subsp. *lactis* CIDCA 133: a health-promoting bacteria. *Probiotics Antimicrob Proteins*. <https://doi.org/10.1007/s12602-021-09826-z>
83. Chokesajjawatee N, Santiyanont P, Chantarasakha K et al (2020) Safety Assessment of a nham starter culture *Lactobacillus plantarum* BCC9546 via Whole-genome Analysis. *Sci Rep* 10:10241. <https://doi.org/10.1038/s41598-020-66857-2>
84. Toropov V, Demyanova E, Shalaeva O et al (2020) Whole-genome sequencing of *Lactobacillus helveticus* D75 and D76 confirms safety and probiotic potential. *Microorganisms* 8:329. <https://doi.org/10.3390/microorganisms8030329>
85. Hummel AS, Hertel C, Holzapfel WH, Franz CMAP (2007) Antibiotic Resistances of starter and probiotic strains of lactic acid bacteria. *Appl Environ Microbiol* 73:730–739. <https://doi.org/10.1128/AEM.02105-06>
86. Anisimova EA, Yarullina DR (2019) Antibiotic resistance of *Lactobacillus* Strains. *Curr Microbiol* 76:1407–1416. <https://doi.org/10.1007/S00284-019-01769-7>
87. Campedelli I, Mathur H, Salvetti E et al (2019) Genus-Wide assessment of antibiotic resistance in *Lactobacillus* spp. *Appl Environ Microbiol*. <https://doi.org/10.1128/AEM.01738-18>
88. Goldstein EJC, Tyrrell KL, Citron DM (2015) *Lactobacillus* Species: taxonomic complexity and controversial susceptibilities. *Clin Infect Dis* 60:S98–S107. <https://doi.org/10.1093/cid/civ072>
89. Das DJ, Shankar A, Johnson JB, Thomas S (2020) Critical insights into antibiotic resistance transferability in probiotic *Lactobacillus*. *Nutrition* 69:110567. <https://doi.org/10.1016/j.nut.2019.110567>
90. Sharma P, Tomar SK, Goswami P et al (2014) Antibiotic resistance among commercially available probiotics. *Food Res Int* 57:176–195. <https://doi.org/10.1016/j.foodres.2014.01.025>

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